

Q1 - 24 June - Shift 2

A circular coil of 1000 turns each with area 1m^2 is rotated about its vertical diameter at the rate of one revolution per second in a uniform horizontal magnetic field of 0.07T . The maximum voltage generation will be _____ V.

*Space for your notes:***Q2 - 25 June - Shift 1**

The current in a coil of self inductance 2.0 H is increasing according to $I = 2\sin(t^2)\text{A}$. The amount of energy spent during the period when current changes from 0 to 2A is _____ J.

*Space for your notes:***Q3 - 26 June - Shift 1**

The magnetic flux through a coil perpendicular to its plane is varying according to the relation $\phi = (5t^3 + 4t + 2t - 5)$ Weber. If the resistance of the coil is 5 ohm , then the induced current through the coil at $t = 2\text{ sec}$ will be:

Space for your notes:

- (A) 15.6 A (B) 16.6 A
(C) 17.6 A (D) 18.6 A

Q4 - 26 June - Shift 1

A 10Ω , 20 mH coil carrying constant current is connected to a battery of 20 V through a switch is opened current becomes zero in $100\mu\text{s}$. The average emf induced in the coil is V.

Space for your notes:

Q5 - 26 June - Shift 2

Two coils of self inductance L_1 and L_2 are connected in series combination having mutual inductance of the coils as M . The equivalent self inductance of the combination will be :



- (A) $\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{M}$ (B) $L_1 + L_2 + M$
(C) $L_1 + L_2 + 2M$ (D) $L_1 + L_2 - 2M$

Space for your notes:

Q6 - 26 June - Shift 2

A metallic conductor of length 1m rotates in a vertical plane parallel to east-west direction about one of its end with angular velocity 5 rad/s . If the horizontal component of earth's magnetic field is $0.2 \times 10^{-4}\text{ T}$, then emf induced between the two ends of the conductor is :

- (A) $5\mu\text{V}$ (B) $50\mu\text{V}$ (C) 5mV (D) 50mV

Space for your notes:

Q7 - 27 June - Shift 2

A metallic rod of length 20 cm is placed in North-South direction and is moved at a constant speed of 20 m/s towards East. The horizontal component of the Earth's magnetic field at that place is $4 \times 10^{-3}\text{ T}$ and the angle of dip is 45° . The emf induced in the rod is _____ mV.

Space for your notes:

Q8 - 29 June - Shift 2

Given below are two statements :

Statement I: The electric force changes the speed of the charged particle and hence changes its kinetic energy: whereas the magnetic force does not change the kinetic energy of the charged particle.

Statement II: The electric force accelerates the positively charged particle perpendicular to the direction of electric field. The magnetic force accelerates the moving charged particle along the direction of magnetic field. In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Both Statement I and Statement II are correct.
- (B) Both Statement I and Statement II are incorrect.
- (C) Statement I is correct but Statement II is incorrect.
- (D) Statement I is incorrect but Statement II is correct.

Space for your notes:

Answer Key

Q1 (440)

Q2 (4)

Q3 (A)

Q4 (400)

Q5 (D)

Q6 (B)

Q7 (16)

Q8 (C)

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Q1 (440)

$$\epsilon_{\max} = BAN\omega$$

$$= 0.07 \times 1 \times 10^3 \times 2\pi$$

$$= 140\pi \approx 440$$

Q2 (4)

$$I = 2 \sin(t^2) \Rightarrow dI = 4t \sin(t^2) dt$$

$$\text{If } I = 0 \Rightarrow t = 0$$

$$\text{and } I = 2 \Rightarrow 2 = 2 \sin t^2$$

$$\Rightarrow t = \sqrt{\frac{\pi}{2}}$$

$$E = \int LI dI$$

$$= \int 2 \times 2 \sin(t^2) \times 4t \cos(t^2) dt$$

$$= 8 \int_0^{\sqrt{\pi/2}} t \sin(2t^2) dt$$

$$= 2 \left[-\cos(2t^2) \right]_0^{\sqrt{\pi/2}}$$

$$= 2 \left[-\cos \pi + \cos 0 \right] = 4$$

Q3 (A)

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$$\phi = 5t^3 + 4t^2 + 2t - 5$$

$$|e| = \frac{d\phi}{dt} = 15t^2 + 8t + 2$$

$$\text{At } t = 2, |e| = 15 \times 2^2 + 8 \times 2 + 2$$

$$\Rightarrow e = 78V \Rightarrow I = \frac{e}{R} = \frac{78}{5} = 15.60$$

Q4 (400)

$$\langle \varepsilon \rangle = \frac{\int \varepsilon dt}{\int dt} = \frac{\int (L di / dt) dt}{\int dt} = \frac{L \int di}{\int dt}$$

$$\langle \varepsilon \rangle = \frac{L \Delta i}{\Delta t}$$

$$i_0 = \frac{V}{R} = \frac{20}{10} = 2A, \text{ if } i = 0A$$

$$T = 100 \mu s, L = 20 \text{ mH}$$

$$\langle \varepsilon \rangle = \frac{20 \times 10^{-3} \times (2 - 0)}{100 \times 10^{-6}}$$

$$= \frac{2 \times 10^3}{5}$$

$$\langle \varepsilon \rangle = 400V$$

Q5 (D)

Current on both the inductor is in opposite direction.

Hence :

$$L_{eq} = L_1 + L_2 - 2M$$

Q6 (B)

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Hints and Solutions

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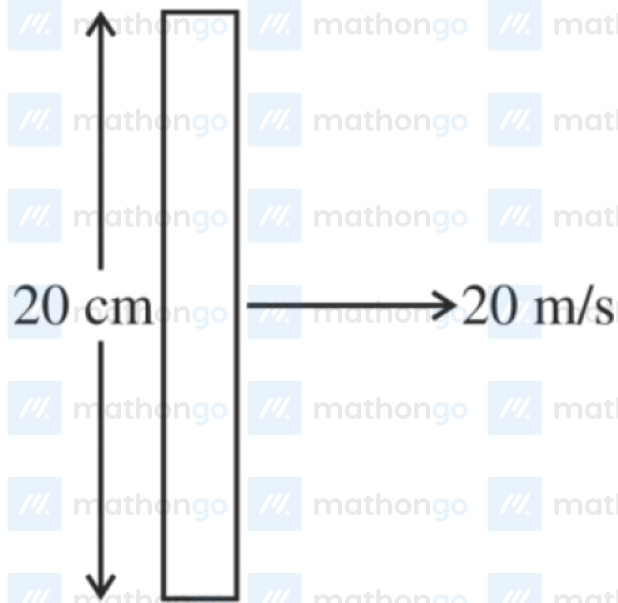
emf induced between the two ends $= \frac{B_H \omega l^2}{2}$

$$\frac{0.2 \times 10^{-4} \times 5 \times 1}{2} = 0.5 \times 10^{-4} = 50 \times 10^{-6} \text{ V} = 50 \mu\text{V}$$

Q7 (16)

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$$B_H = 4 \times 10^{-3} \text{ T}$$

$$\theta \rightarrow 45^\circ$$

$$B_V = B_H$$

$$\epsilon = (\vec{V} \times \vec{B}) \cdot \vec{\ell}$$

$$= ((4 \times 10^{-3})(20)) \frac{20}{100}$$

$$= 16 \times 10^{-3} \text{ V} = 16 \text{ mV}$$

Q8 (C)

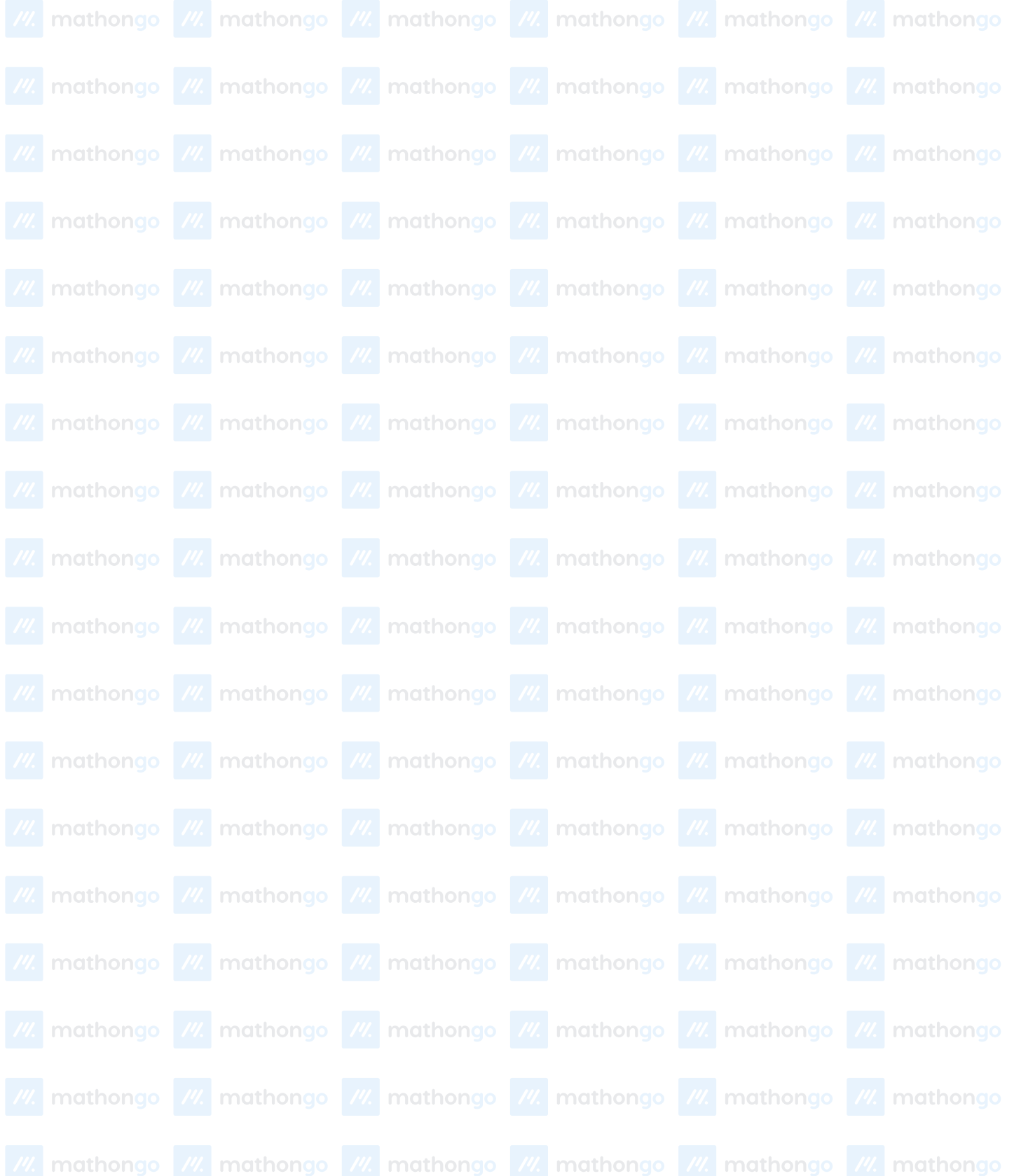
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Hints and Solutions

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Electric field can change speed and kinetic energy
but magnetic field can not change speed Δ KE.

Because magnetic force is always $\perp$ to velocity.



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